

Final Project Summary

Project Title: Orange Retail

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Links: [GitHub Repository](#)

[YouTube Screencast](#)

Orange Retail is a Laravel MVC supermarket web application developed for the Server-Side Development module. The final system supports both customer shopping tasks and administrator management tasks in one structured platform. The application was built to solve a common problem in small and medium supermarkets where product information, stock control, and customer orders are often handled through disconnected tools or manual processes.

Current State of the Application

The application is fully structured using the Laravel Model-View-Controller architecture. It includes database-driven functionality, authentication, authorization, validation, and multiple interactive views for both customers and administrators.

Implemented Features

Customer-side features currently implemented:

- User registration and login
- Public product catalogue
- Product search
- Favourite products / saved products
- Checkout process
- Order creation
- Order details page
- Customer dashboard
- Product detail pages
- Category and subcategory browsing
- Shopping cart
- Delivery information form validation
- Order history

Administrator-side features currently implemented:

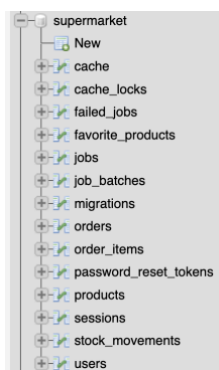
- Admin dashboard
- Product management
- Full CRUD functionality for products
- Stock Centre / inventory management
- Minimum stock level tracking
- Stock alert workflows
- Admin order queue
- Order status updates

MVC and Database

The system follows the Laravel MVC pattern:

- Models manage business entities such as users, products, orders, order items, stock movements, and favourite products.
- Controllers handle requests, validation, business rules, and application flow.
- Blade views display the customer and admin interfaces.

The application uses a relational database and Laravel migrations.



The main database entities in the final project are:

- users

id	name	email	email_verified_at	password	role	remember_token	created_at	updated_at
3	Nikita Smiichyk	nikita@gmail.com	2026-04-04 23:35:02	\$2y\$12\$a33Xsa3Zh/9	admin	NClkQO2VDXCn9V05qz3c4:	2026-03-29 00:45:51	2026-04-04 22:35:02
4	Store User	user@supermarket.test	2026-04-04 22:35:02	\$2y\$12\$rqLMTaJ7yF-	user	NULL	2026-03-29 00:45:51	2026-04-04 22:35:02
5	Hanna Bokariuk	hanna.user@gmail.com	2026-04-04 23:54:52	\$2y\$12\$mcHtwH.JxRl	user	NULL	2026-04-05 02:34:08	2026-04-05 02:34:08

- products

id	name	brand	sku	category	subcategory	unit_type	pack_size	price_value	unit_price_display	stock	minimum_stock_level	is_active	last_restocked_at	next_deliver
960	Kellogg's Crunchy Nut Breakfast Cereal 460g	Kellogg's	KEL-0190-KELL-S-CRUI	Food	Cereals	weight	460g	3.50	€7.61/kg	233	67	1	2026-04-05 20:55:42	2026-04-12 2
989	Sparkling Diet Lemonade 2 Litre Bottle	Daily Sip	DAI-0209-SPAF-DIET-L	Drinks	Fizzy & Soft Drinks	volume	2Litre	0.80	€0.40/litre	217	69	1	2026-04-01 20:55:42	2026-04-13 2
	Heinz Beanz		HEI-0182		Tins.									

- orders

id	user_id	order_number	status	customer_name	customer_email	shipping_address_line_1	shipping_address_line_2	shipping_city	shipping_county	shipping_postal_code	notes	item_count	subi
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- *order_items*

id	order_id	product_id	product_name	product_sku	product_brand	quantity	unit_price	line_total	created_at	updated_at
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- *stock_movements*

id	product_id	user_id	type	quantity_change	note	occurred_at	created_at	updated_at
1	893	3	schedule_update	0	NULL	2026-04-05 19:36:22	2026-04-05 19:36:22	2026-04-05 19:36:22
2	858	3	manual_decrease	-20	NULL	2026-04-06 20:45:52	2026-04-06 20:45:52	2026-04-06 20:45:52
3	858	3	manual_decrease	-1	NULL	2026-04-06 20:45:56	2026-04-06 20:45:56	2026-04-06 20:45:56
4	858	3	restock	10	NULL	2026-04-06 20:46:02	2026-04-06 20:46:02	2026-04-06 20:46:02

- *favorite_products*

id	user_id	product_id	created_at	updated_at
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Authentication and Authorization

The project includes user authentication with registration, login, logout, and password hashing. Authorization is role-based. Customers can access shopping features such as cart, checkout, favourites, and order history, while admin-only routes protect product management, stock management, and order management features.

Validation

Both client-side and server-side validation are implemented.

Examples include:

- *required login and registration fields*
- *email validation*
- *positive numeric price validation*
- *non-negative stock validation*
- *checkout form validation*
- *stock-aware ordering rules*
- *protected form submissions with CSRF handling*

CRUD Functionality

The main fully implemented CRUD feature is product management.

Administrators can:

- *create products*
- *view products*
- *update products*
- *delete products*

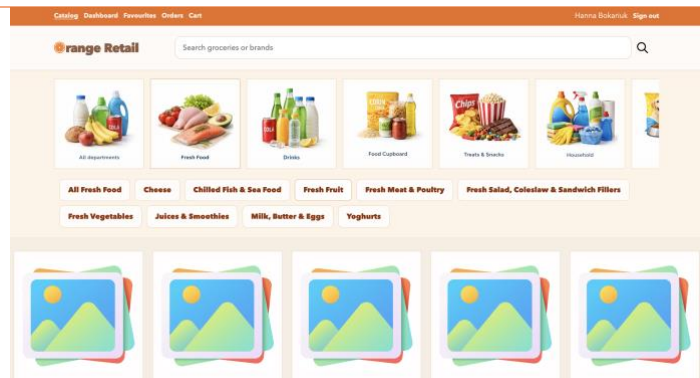
This satisfies the required CRUD component of the assignment.

Views and Pages

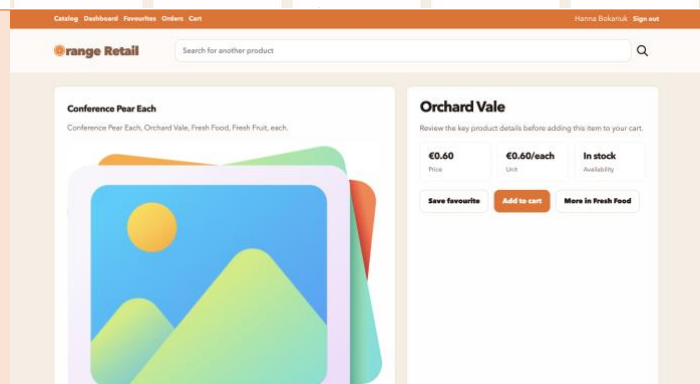
The project contains more than the minimum five required views.

Key pages include:

- **homepage / catalogue**



- **product details page**



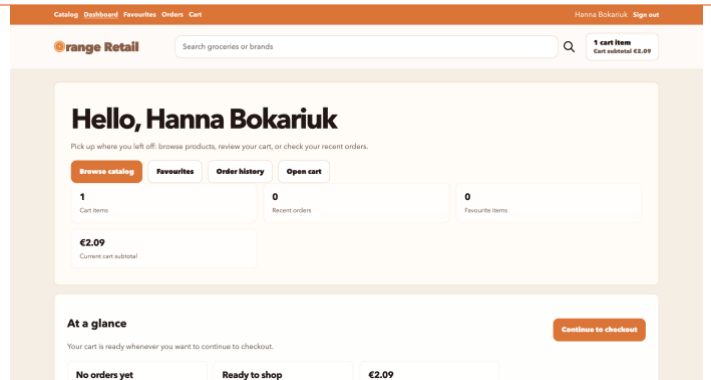
- **login page**

The screenshot shows the login page. It has a navigation bar at the top with links: Catalog, Dashboard, Favorites, Orders, Cart. On the right, it says 'Hanna Bokanuk Sign out'. Below the navigation bar is a search bar with the placeholder text 'Search for another product'. The main content area is a white box with the title 'Login' and a subtitle 'Enter your email and password to continue.' Below this are two input fields: 'Email' and 'Password'. There is a checkbox labeled 'Remember me' and two buttons: 'Login' and 'Register'.

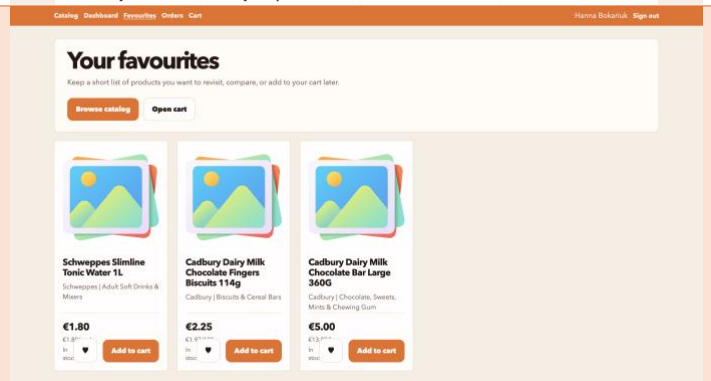
- **registration page**

The screenshot shows the registration page. It has a navigation bar at the top with links: Catalog, Dashboard, Favorites, Orders, Cart. On the right, it says 'Hanna Bokanuk Sign out'. Below the navigation bar is a search bar with the placeholder text 'Search for another product'. The main content area is a white box with the title 'Register' and a subtitle 'Create a new account to access the supermarket catalog.' Below this are four input fields: 'Full name', 'Email', 'Password', and 'Confirm password'. There is a small text 'Please fill in this field.' below the 'Confirm password' field. At the bottom, there are two buttons: 'Create account' and 'Login'.

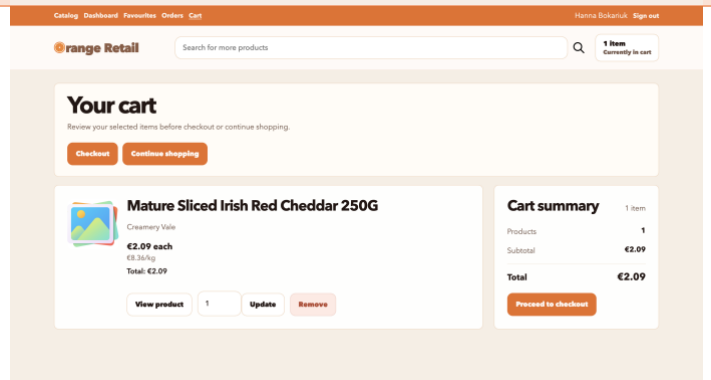
○ **customer dashboard**



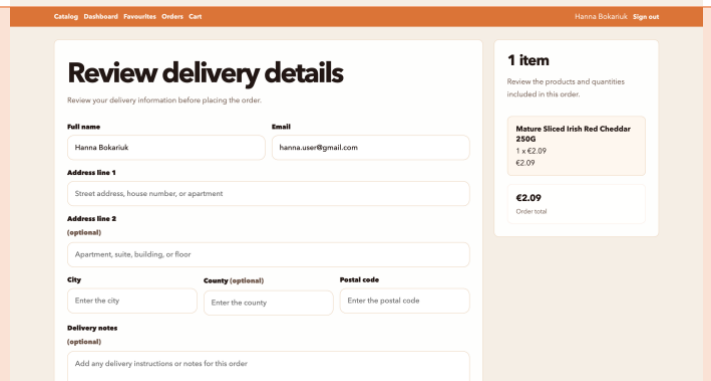
○ **favourites page**



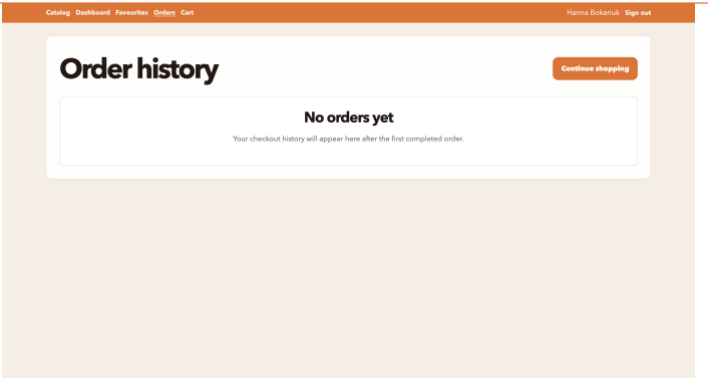
○ **shopping cart**



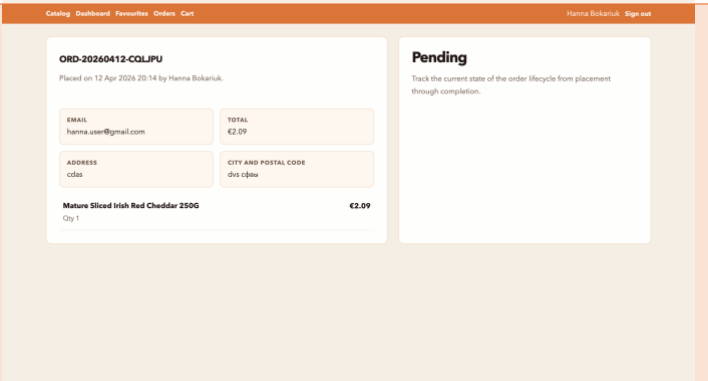
○ **checkout page**



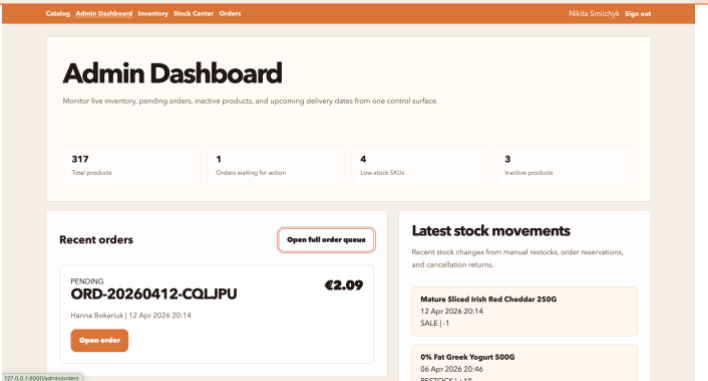
○ **order history page**



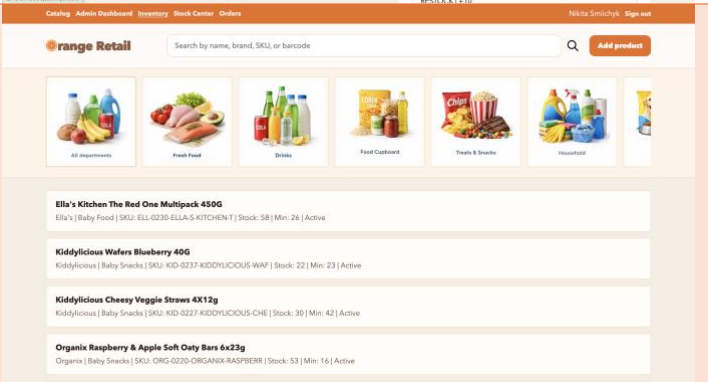
○ **order details page**

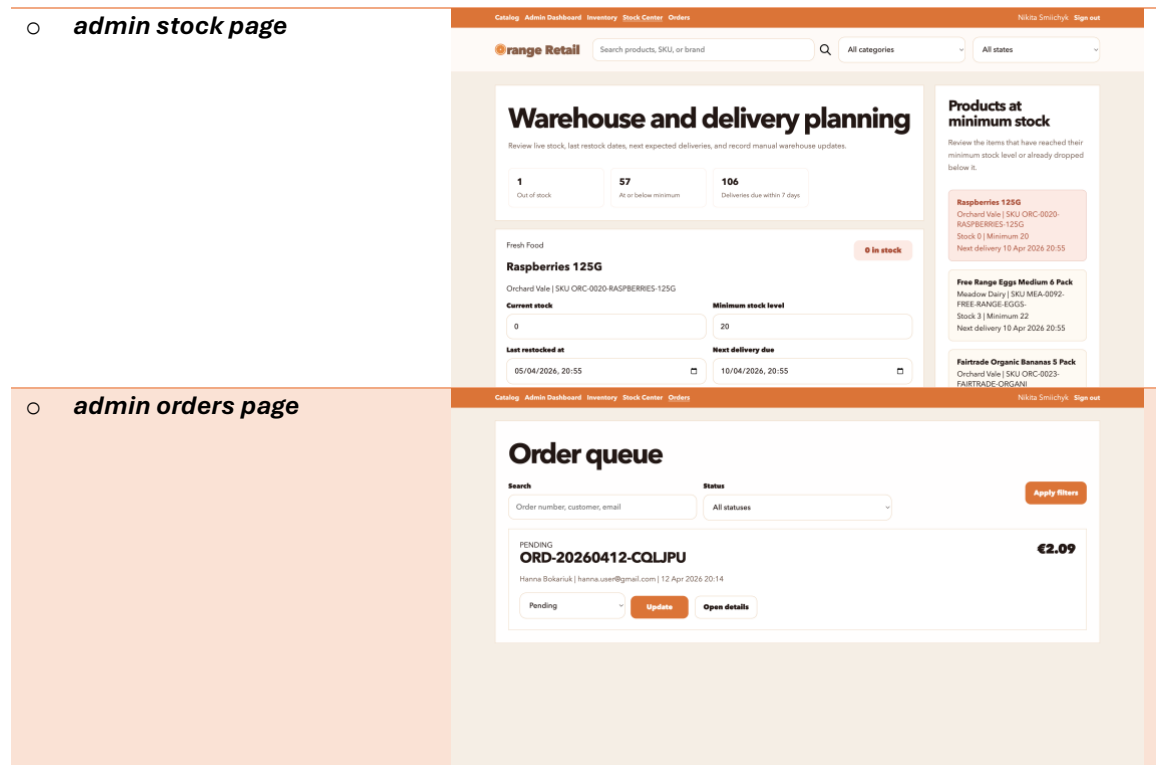


○ **admin dashboard**



○ **admin products page**





Git and Development Process

The project is stored in a GitHub repository and shows regular development activity through meaningful commits. The repository also contains merge history and reflects ongoing improvements to features, validation, UI structure, and workflow behaviour during development.

Assumptions and Limitations

The project is designed for a local development environment. Payment is simulated and not connected to a real payment provider. The system does not include delivery route management, advanced analytics, or multilingual support. Some products use placeholder images where real product images are not available.

Conclusion

The current project meets the core requirements of the assignment by delivering a working Laravel MVC application with authentication, authorization, relational database support, migrations, validation, multiple views, and a fully implemented CRUD feature. In addition, the final system includes extended functionality such as favourites, stock alerts, shopping cart flow, checkout, and order management.